

Instructions for TMA:

- i. Answers must be written by hand.
- ii. Spiral binding is enough.
- iii. Do not copy from others.
- iv. Keep the TMA standard and simple.
- v. The submission deadline is the last class - 01 August, 2025.
- vi. Submit to Ali Vy.

TMA Questions:**TMA - Computer Basics**

1. What is Computer? Write the characteristics of first-generation Computer.
2. Briefly describe the computer generations.
3. How 625.536 and 735 can be express in expanded notation in respect of decimal number system?
4. Convert the following numbers as directed.
 - i. $(110101)_2 = (?)_{10}$
 - ii. $(73D5)_{16} = (?)_{10}$
 - iii. $(456)_8 = (?)_2$
5. Evaluate the following:
 - i. $10110 + 1100$
 - ii. 1011×101
 - iii. $10010 - 1011$
 - iv. $11010011 \div 11$
 - v. $11011 - 10010$ (using 2's complement)
6. Explain the operation of primary logic gates.
7. Draw the diagram and write truth table of AND and XOR gate.
8. Describe the function of the control unit.
9. Discuss the port and interfaces of computer.
10. Write the purpose of Scanning device.
11. Describe difference primary storage of computer systems.
12. What is the difference between system software and application software?
13. What is the purpose of a compiler in programming?
14. "Hardware is useless without software." Briefly Explain.
15. Distinguish between batch processing and multiprogramming.
16. Explain the responsibilities of a system analyst in a computer department.
17. What is the function of web browser?
18. Discuss about OSI layers.

Instructions for the Record Book:

- i. Answers must be written by hand.
- ii. Spiral binding is enough.
- iii. Do not copy from others.
- iv. Keep the TMA standard and simple.
- v. The submission deadline - practical examination day 1201.
- vi. Submit to Ali Vy before entering practical examination.

Record Book Questions:

1. Observation and identification of ports and interfacing of PC.
2. Installation and assembly of a PC.
3. Installation, removal and study of Hard disk or CD ROM
4. Study of the tools in the control panel of Windows.

Follow the format for each lab mentioned below:

- i. Title
- ii. Objectives
- iii. Theory
- iv. Working diagrams
- v. Apparatus
- vi. Procedure
- vii. Results
- viii. Discussion and conclusion